

5. Minimal Spanning Tree (M.S.T.)

Suppose there are n cities to be connected by roads or by railways or by air. Suppose the cost of connecting two cities v and v' is $C(v, v')$. The problem is to find the way in which the cities should be connected, so that the cost is minimum. This is known as **"the connector problem"**.

The connector problem can be recast into a problem of graph theory as follows. Let the cities be denoted by the vertices $v_1, v_2, v_3, \dots$. If two cities v and v' are to be connected then we say that there is an edge e between v and v' . If these edges are $e_1, e_2, e_3, \dots$ then we have a graph $G = (V, E)$. Let $C(e)$ denote the cost of constructing the road e . The connector problem now becomes the problem of finding a spanning tree of the graph G such that the total cost of

constructing the roads is minimum. i.e. to find the spanning tree of $G = (V, E)$ such that $\sum C(e)$ is minimum where summation runs over all the edges of the spanning tree.

Definition : Let $G = (V, E)$ be a connected graph. Let $e \in E$ and $C(e)$ be a non-negative number associated with e . It is called the **cost** or the **weight** of e . When costs or weights thus are associated with all edges the graph is called the **weighted graph**. A spanning tree T of the graph G such that the total cost $\sum C(e)$ is minimum is called a **minimal spanning tree (MST)** of the graph G . The sum $\sum C(e)$ for the minimal spanning tree is called the **minimum cost** or the **minimum weight** of the graph G .

There are various algorithms to find MST and the weight of G . We shall study one of them, known as **Kruskal's algorithm**.

6. Kruskal's Algorithm

Suppose we are given a connected weighted graph. First of all name all the vertices. Also list all the edges in the increasing order of their weights (i.e. taking the minimum cost first).

Now, jot down all the vertices along with their names given in the graph (See Examples).

Step 1 : Get the first edge by joining the vertices topping the above list.

Step 2 : Get the second edge by joining the vertices second in the list.

Step 3 : Consider the next (3rd) edge in the list. If it does not form a cycle with already selected edges then join the vertices to get the edge. If it forms a cycle with the already selected edges, reject it.

Step 4 : Repeat the above step till you get a spanning tree i.e. you have joined all the vertices.

This is the required **Minimal Spanning Tree**.

Example 1 : Find the spanning tree of the graph shown in Fig. 9.7 with minimum weight.

Solution : (i) We first jot down all the vertices as shown in Fig. 9.7 (a).

(ii) We now list the edges in increasing order of their costs (weights).

(1) $\{A, B\}$, (2) $\{D, E\}$, (3) $\{E, C\}$,

(4) $\{C, B\}$, (5) $\{B, D\}$, (6) $\{A, C\}$

Step 1 : We obtain the first edge by joining the vertices topping the above list. We join the vertices A and B . [Fig. 9.7 (b)]

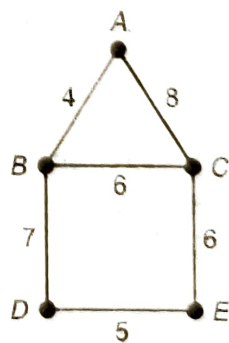


Fig. 9.7

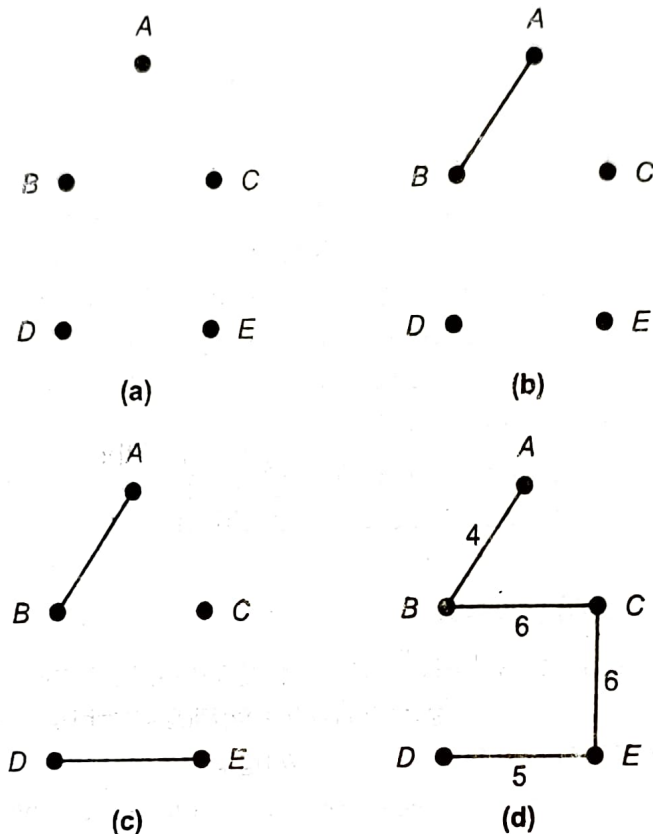


Fig. 9.7

Step 2 : We obtain the second edge by joining the vertices second in the list. We join the vertices D and E . [Fig. 9.7 (c)]

Step 3 : We obtain the third edge by joining the vertices third in the order. There are two pairs of vertices having weight 6. [Fig. 9.7 (d)]

Since they do not form a cycle, we join both these pairs. We thus get the edges EC and CB .

Since all the vertices are now joined, this is the required minimum spanning tree. By adding the costs, we get the minimal cost $4 + 5 + 6 + 6 = 21$.

Example 2 : Find the minimal spanning tree of the graph shown in Fig. 9.8.

Sol. : (i) We first jot down all the vertices as shown in Fig. 9.8 (a).

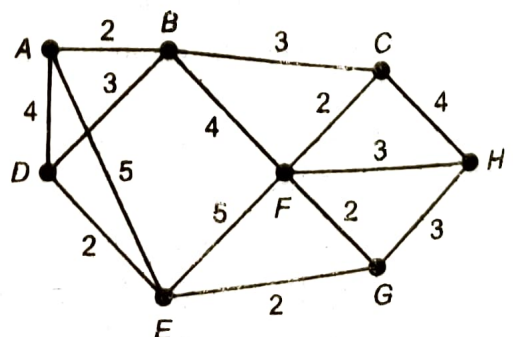


Fig. 9.8

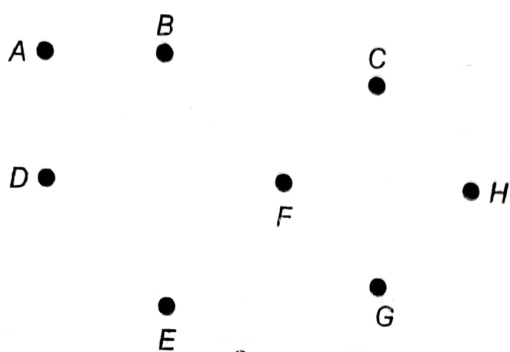


Fig. 9.8 (a)

(ii) We now list the edges in increasing order of their costs (weights).

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| (1) $\{A, B\}$, | (2) $\{D, E\}$, | (3) $\{E, G\}$, | (4) $\{F, C\}$, |
| (5) $\{F, G\}$, | (6) $\{B, C\}$, | (7) $\{D, B\}$, | (8) $\{F, H\}$, |
| (9) $\{G, H\}$, | (10) $\{A, D\}$, | (11) $\{C, H\}$, | (12) $\{B, F\}$, |
| (13) $\{A, E\}$, | (14) $\{E, F\}$. | | |

Step 1 : We obtain the first edge by the vertices topping the list which do not form a cycle. Hence, we join the following pairs of vertices (A, B) , (D, E) , (E, G) , (F, C) and (F, G) . [Fig. 9.8 (b)]

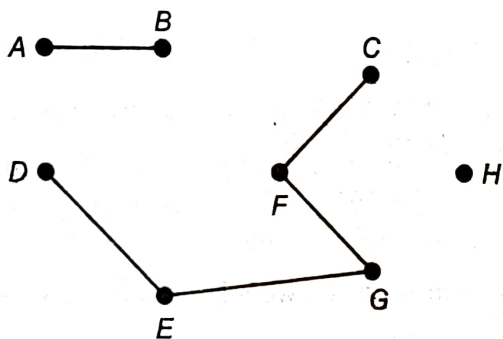


Fig. 9.8 (b)

Step 2 : We join the vertices having the next higher weight viz. 3, if they do not form a cycle. We join B and C and F and H . [Fig. 9.8 (c)]

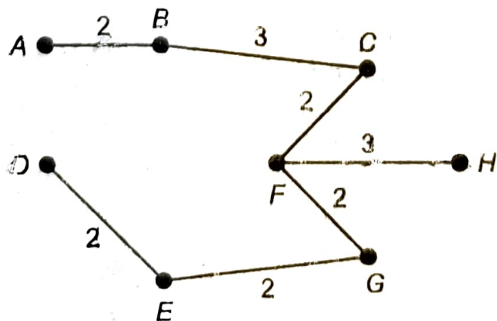


Fig. 9.8 (c)

Since all vertices are connected, this is the minimum spanning tree.

By adding the cost, we get the minimal cost $2 + 2 + 2 + 2 + 2 + 3 + 3 = 16$.

Example 3 : Obtain minimal spanning tree and its weight for the graph shown in Fig. 9.9.

Sol. : We first name the vertices and jot down all of them as shown below. [Fig. 9.9 (b)]

We now list the edges in the increasing order of their costs.

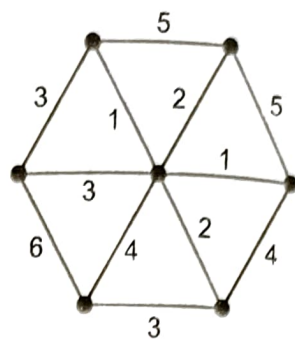


Fig. 9.9

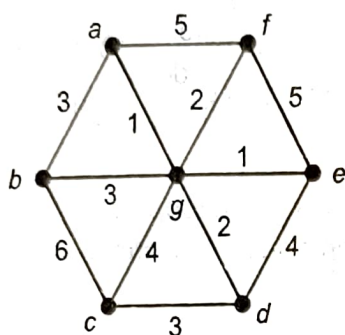


Fig. 9.9 (a)

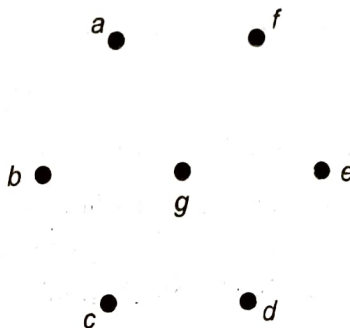


Fig. 9.9 (b)

Since the minimum cost is one we start with the edge $\{a, g\}$.

(1) $\{a, g\}$, (2) $\{g, e\}$, (3) $\{g, f\}$, (4) $\{g, d\}$, (2) $\{g, b\}$, (6) $\{b, a\}$,

(7) $\{c, d\}$, (8) $\{c, g\}$, (9) $\{d, e\}$, (10) $\{e, f\}$, (11) $\{f, a\}$, (12) $\{b, c\}$.

Step 1 : Since there are two pairs of vertices $\{a, g\}$ and $\{g, e\}$ with minimum cost one, we join them.

Step 2 : The two pairs of vertices having weight two are $\{g, f\}$ and $\{g, d\}$. Since they do not form cycles with the first two edges, we join them.

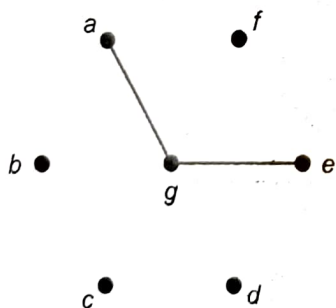


Fig. 9.9 (c)

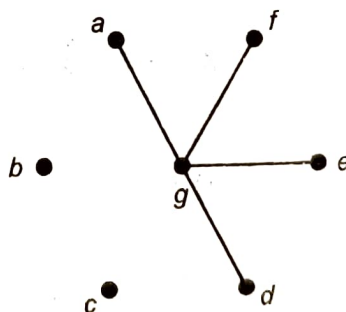


Fig. 9.9 (d)

Step 3 : There are three pairs of vertices having weight three. They are $\{g, b\}$, $\{b, a\}$, $\{d, c\}$. We join the vertices g, b and d, c . But we do not join the vertices b, a because this edge forms a cycle. So we reject this pair of vertices.

Since all the vertices are now joined this is the required minimum spanning tree. By adding the costs we get, the minimal cost $1 + 1 + 2 + 2 + 3 + 3 = 12$.

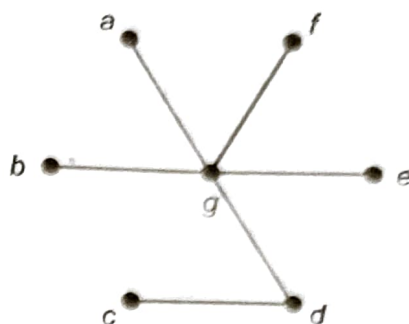


Fig. 9.9 (e)

Remark

When there are more than one pairs of vertices having the same cost, we can choose any one of them. Of course care must be taken to see that this edge does not form a cycle because a tree by definition does not have a cycle.

Example 4 : Obtain the minimum spanning tree for the following graph.

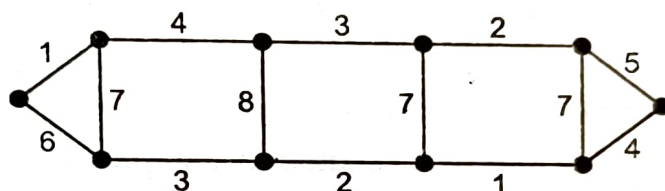


Fig. 9.10

Sol. : Name the vertices as $a, b, c, \dots, j$ and jot down them as shown in Fig. 9.10(b).

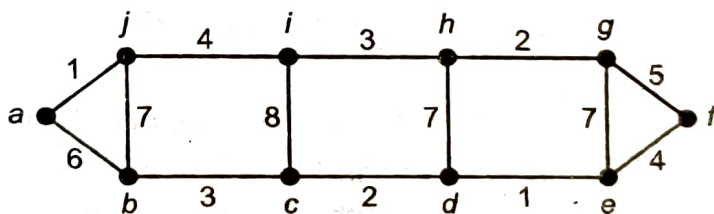


Fig. 9.10 (a)

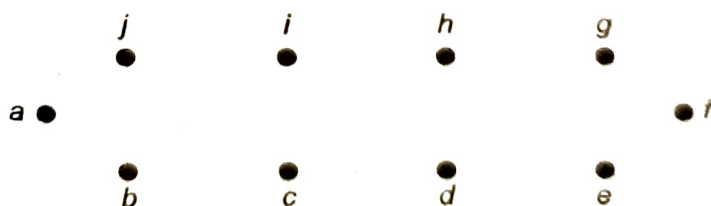


Fig. 9.10 (b)

Prepare the list of edges in increasing order of their costs.

- (1) $\{a, j\}$, (2) $\{d, e\}$, (3) $\{c, d\}$, (4) $\{h, g\}$, (5) $\{b, c\}$,
 (6) $\{i, h\}$, (7) $\{e, f\}$, (8) $\{i, j\}$, (9) $\{f, g\}$, (10) $\{a, b\}$,
 (11) $\{b, j\}$, (12) $\{d, h\}$, (13) $\{e, g\}$, (14) $\{c, i\}$.

Step 1 : We join the vertices in the increasing order of weights and get the first nine edges since these do not form a cycle.

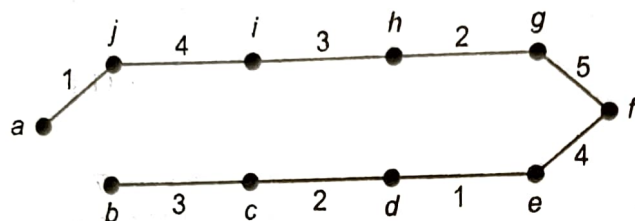


Fig. 9.10 (c)

Step 2 : Since all the vertices are joined this is the required minimum spanning tree.

By adding the costs, we get

$$\text{The minimal cost} = 1 + 4 + 3 + 2 + 5 + 4 + 1 + 2 + 3 = 25$$

7. Kruskal's Second Method

Kruskal's algorithm can be executed in the reverse order also. The following steps are involved in this method.

First draw the given graph with the weights of the edges.

List all the edges in decreasing order of costs.

Step 1 : Find the edge which has maximum cost and which forms a cycle. Delete this edge say e_1 and get the graph $G' = G \sim e_1$. If G' is a tree the problem is solved. If G' is a cycle go to the step two.

Step 2 : Find an edge which has maximum weight in G' and which forms a cycle in G' . Delete this edge say e_2 and get the graph $G'' = G' \sim e_2$. If G'' is a tree the problem is solved. If G'' is not a tree repeat the steps till you get M.S.T.

In practice the problem can be solved easily and quickly like this. Draw the graph showing the weight of each edge. Now, starting from the highest weight go on deleting the edges (crossing them) which form cycle, till you get a tree.

We solve the above two problems (3) and (4), page 9-13 and 9-14 below again by this method.

Example 3 : We cross the edges of weights 6, 5, 5, 4, 4, 3 in that order to break the cycle at each stage. We thus get the M.S.T. Fig. 9.11 (b).

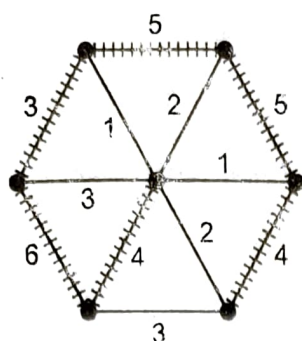


Fig. 9.11 (a)

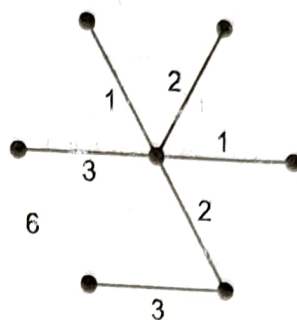


Fig. 9.11 (b)

Example 4 : We cross the edges of weights 8, 7, 7, 7, 6 in that order to break the cycle at each stage. We thus, get the M.S.T. Fig. 9.12 (b).

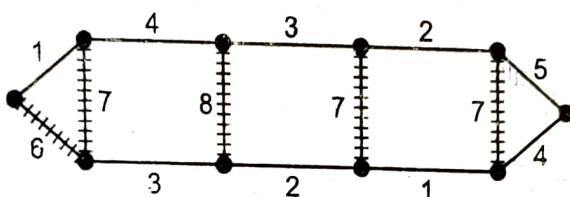


Fig. 9.12 (a)

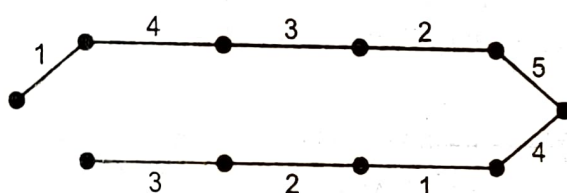


Fig. 9.12 (b)

Example 5 : Find minimal spanning tree for the graph shown in Fig. 9.13.

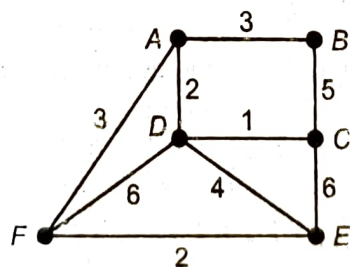


Fig. 9.13

Sol. : We shall follow the second method.

Delete the edge $\{C, E\}$ whose weight is 6. Delete the edge $\{F, D\}$ whose weight is 6. Delete edge $\{B, C\}$ whose weight is 5. Delete edge $\{D, E\}$ whose weight is 4.

The resulting MST is shown in Fig. 9.13 (b).

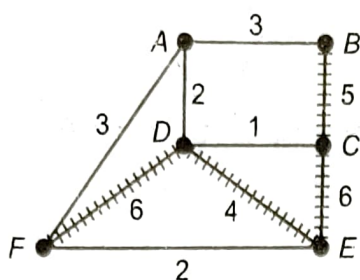


Fig. 9.13 (a)

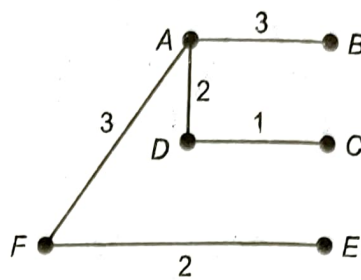


Fig 9.13 (b)

The minimal cost = $3 + 3 + 2 + 2 + 1 = 11$.

Example 6 : Obtain MST and its weight for the graphs shown in Fig. 9.14.

Sol. : The edges in decreasing order of costs are

- (1) $\{c, d\}, 8,$ (2) $\{g, d\}, 7,$ (3) $\{d, e\}, 6,$
- (4) $\{g, e\}, 6,$ (5) $\{f, e\}, 6,$ (6) $\{g, b\}, 4,$
- (7) $\{g, c\}, 3,$ (8) $\{g, f\}, 3,$ (9) $\{a, b\}, 3,$
- (10) $\{g, a\}, 2,$ (11) $\{a, f\}, 2,$ (12) $\{b, c\}, 1.$

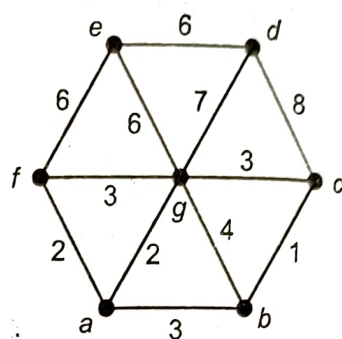


Fig. 9.14

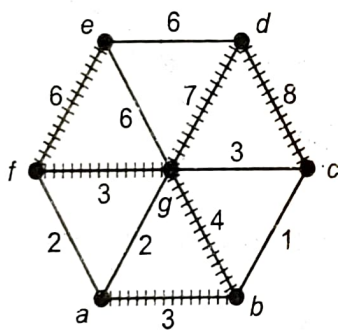


Fig. 9.14 (a)

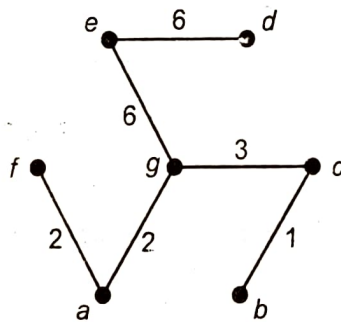


Fig. 9.14 (b)

We shall delete the edges in decreasing order of weights which form cycles. First we delete the edge $\{d, c\}$ which forms a cycle. We then delete the edge $\{g, d\}$ which also forms a cycle. We then delete the edge $\{f, e\}$ which forms a cycle. We then delete $\{g, b\}$ which forms a cycle. The edges $\{g, f\}$ and $\{a, b\}$ also form cycles. Hence, are deleted.

Thus, we get the MST as shown in Fig. 9.14 (b).

The minimal cost = $6 + 6 + 3 + 2 + 2 + 1 = 20$.

8. Prim's or Disjkstra's Algorithm

Another way to compute minimum spanning tree for a weighted graph is **Prim's or Disjkstra's Algorithm**. It was discovered by Prim and Dijkstra simultaneously but independently.

Step 1 : Select any arbitrary vertex say v_0 as tree root.

Step 2 : Find edge $e_1 = (v_0, v_1)$ in E such that its one end vertex v_0 is in T and its weight is minimum.

Join the vertex v_0 to v_1 .

Step 3 : Select the next edge $e_i = (v_i, v_j)$ such that its one end vertex v_i is in T and the other end vertex v_j is not in T . (e_i should not form a circuit) and also the weight of the edge e_i is as small as possible.

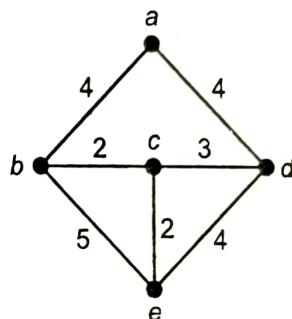
Join the edge e_i and the vertex v_j to T .

Step 4 : Repeat the step 3 until all the vertices of G are in T .

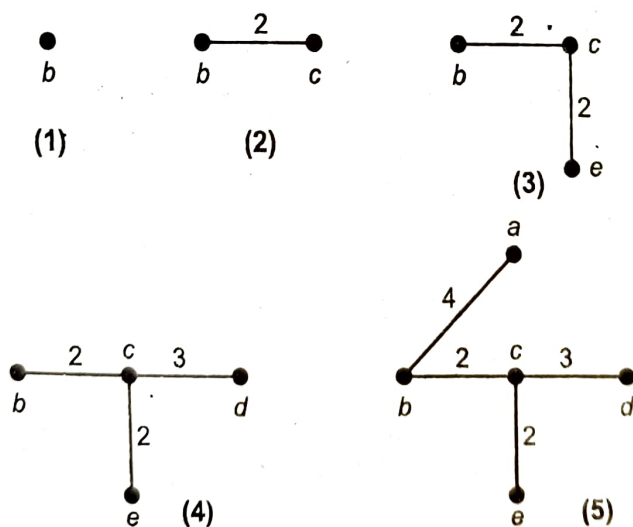
T is the required minimum spanning tree.

Example 1 : Find the minimal spanning tree for the adjoining graph using Prim's algorithm.

Sol. : Starting from b the minimum spanning tree of the given graph can be obtained using Prim's algorithm is given below.



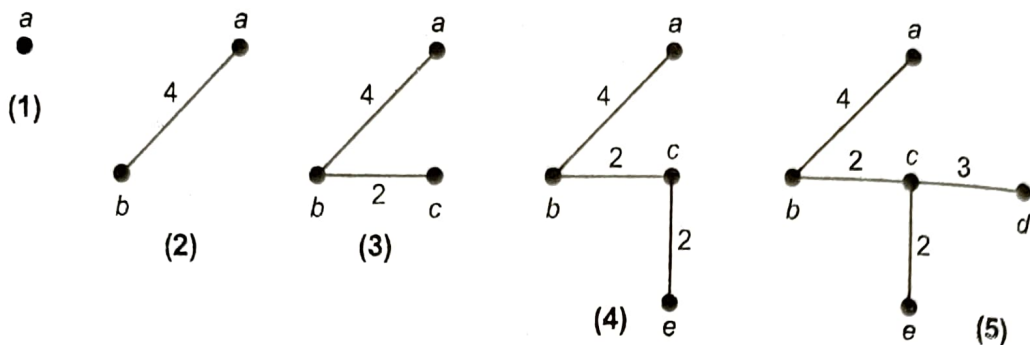
1. We shall start with the vertex b .



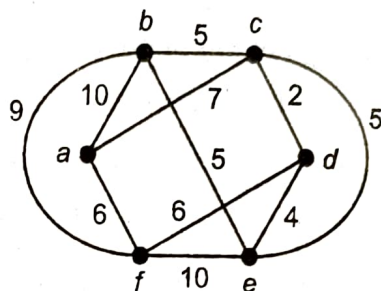
The graph obtained in the last step 5 is a minimum spanning tree.

Its weight = $4 + 2 + 3 + 2 = 11$.

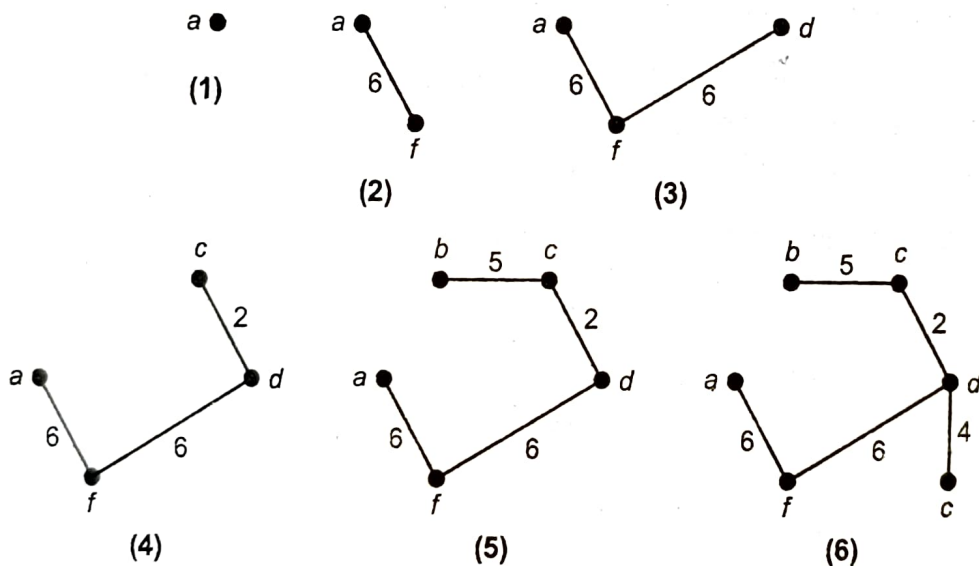
2. Alternatively we may start from any other vertex, say, a . The spanning tree is obtained as follows.



Example 2 : Using Prim's algorithm obtain the minimum spanning tree for the graph given on the right.



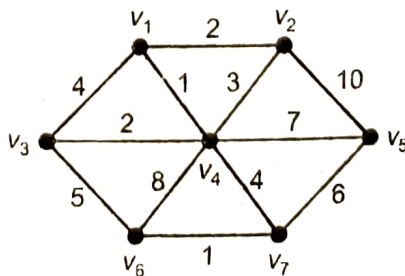
Sol. : Starting from a , we get

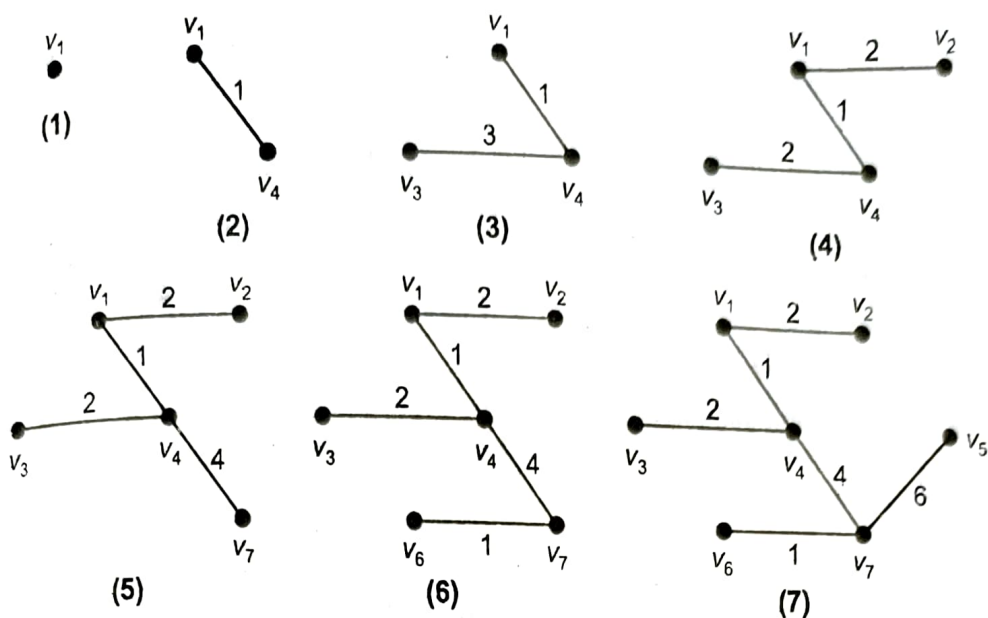


$$\text{Weight} = 6 + 6 + 2 + 5 + 4 = 23.$$

Example 3 : Find the minimum spanning tree using Prim's algorithm for the graph.

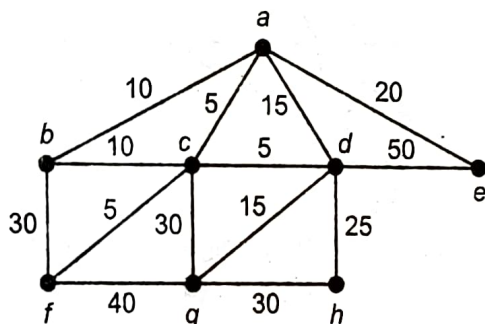
Sol. : Starting from the vertex v_1 ,



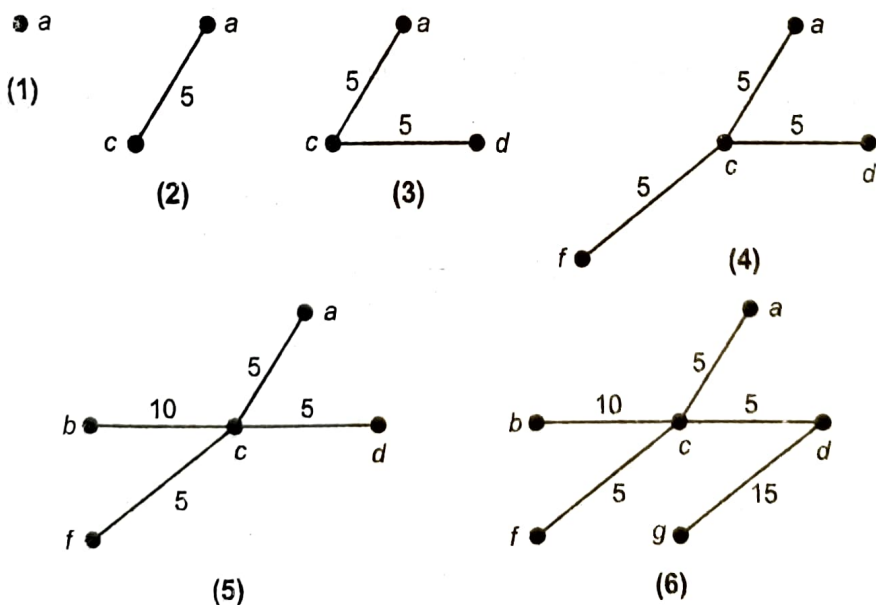


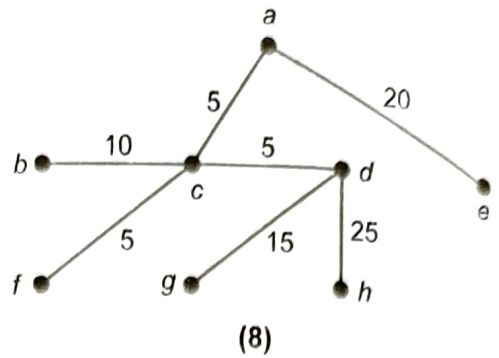
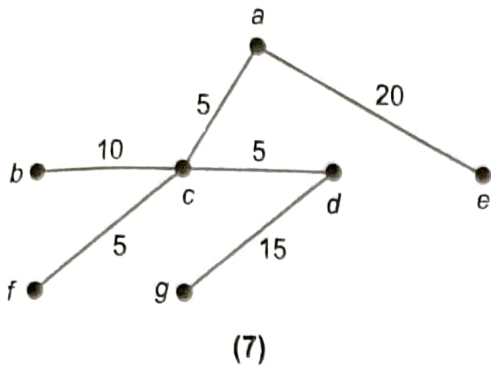
Total cost = $1 + 2 + 2 + 4 + 1 + 6 = 16$.

Example 4 : Use Prim's and Kruskal's algorithm to find minimum spanning tree for the following graph. (M.U. 2008)



Sol. : (a) Prim's Algorithm : We shall start with the vertex a .

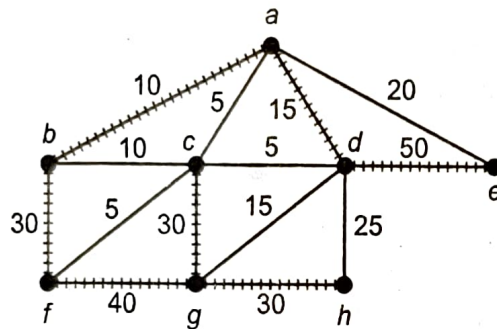




Cost of the minimum spanning tree

$$= 5 + 5 + 5 + 10 + 15 + 25 + 20 = 85.$$

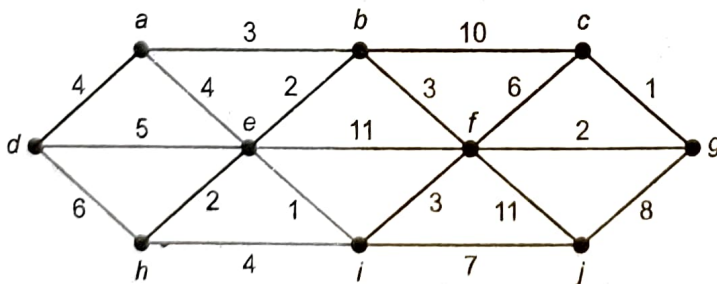
(b) Kruskal Second Method : We shall use Kruskal's second method of deleting edges with highest costs serially in the descending order and breaking the cycles.



We delete the edges of maximum weights and also break the cycles. We delete the edges (d, e) , (f, g) , (g, h) , (c, g) , (a, d) and (a, b) .

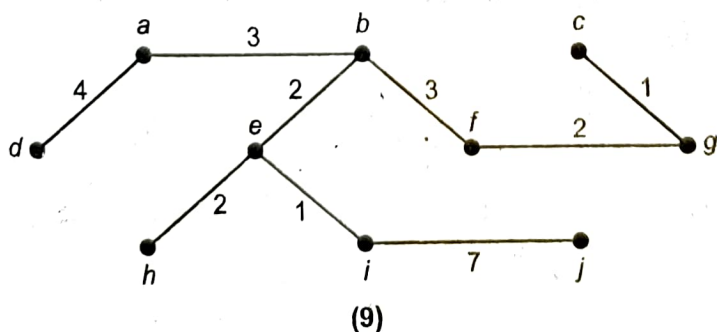
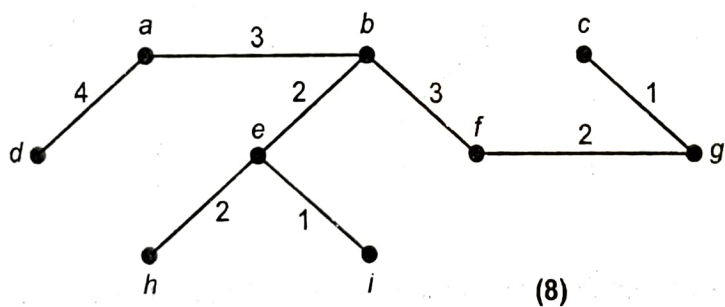
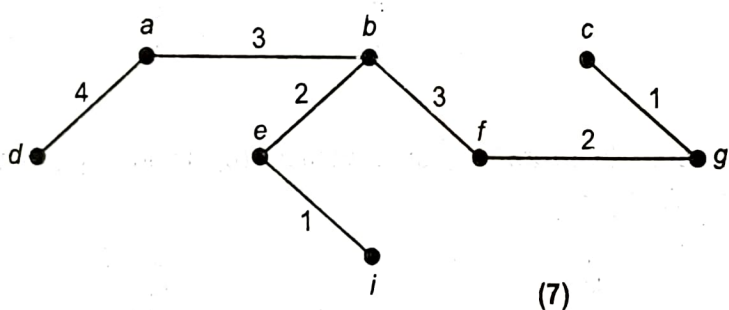
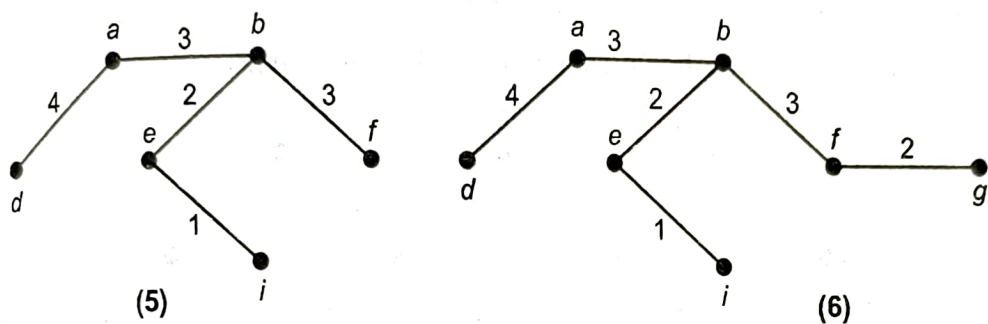
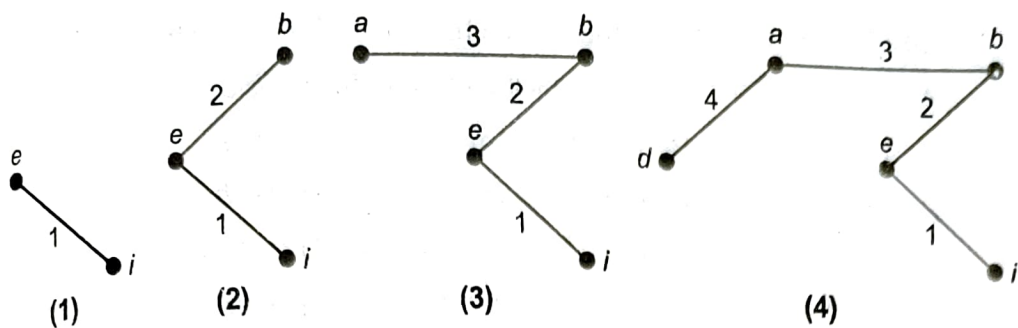
We get the same tree with the same minimum cost as before.

Example 5 : Find the minimum spanning tree using any algorithm for the following graph. (M.U. 2003)



Sol. : (a) By Prim's Algorithm

We start with the vertex e .



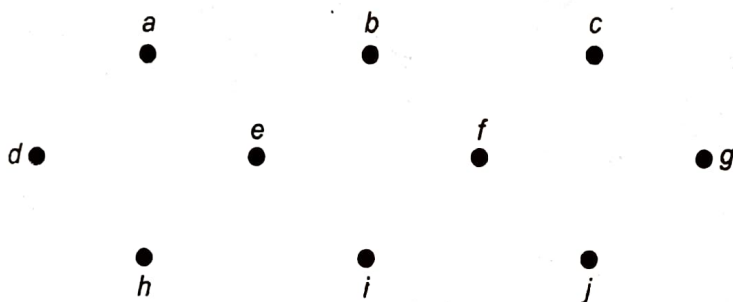
$$\text{Total cost} = 3 + 4 + 2 + 2 + 1 + 7 + 3 + 2 + 1 = 25.$$

(b) By Kruskal's algorithm : We shall follow the first method here.

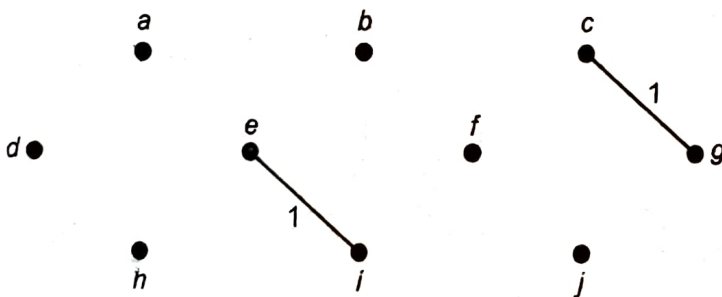
We list the edges in the increasing order of their costs.

- | | | | |
|-----------------|-----------------|-----------------|-----------------|
| (1) (e, i) , | (2) (c, g) , | (3) (e, h) , | (4) (f, g) , |
| (5) (e, b) , | (6) (f, b) , | (7) (f, i) , | (8) (a, b) , |
| (9) (e, a) , | (10) (h, i) , | (11) (a, d) , | (12) (e, d) , |
| (13) (d, h) , | (14) (f, c) , | (15) (i, j) , | (16) (j, g) , |
| (17) (c, b) , | (18) (f, j) , | (19) (e, j) . | |

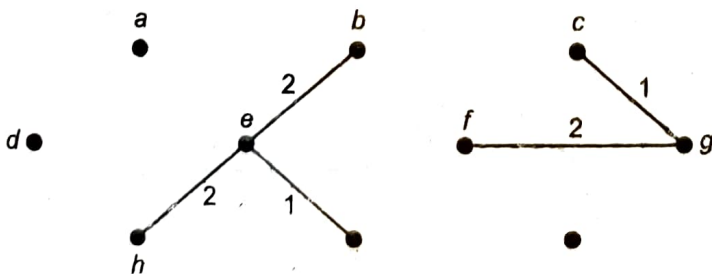
We first jot down the vertices along with their names.



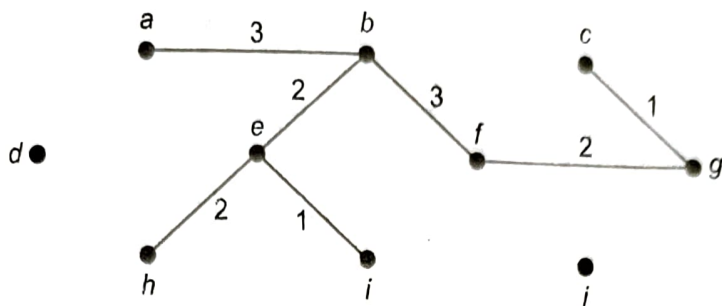
Step 1 : Since there are two pairs of vertices (e, i) and (c, g) with minimum cost one, we join them.



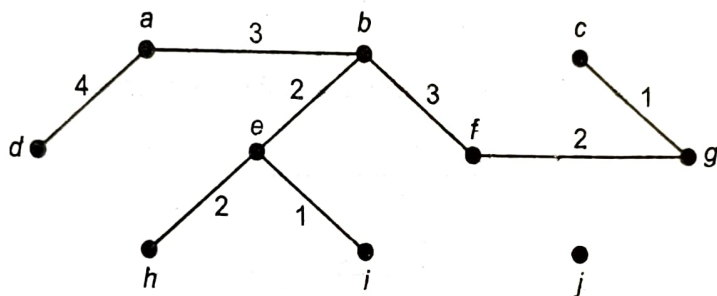
Step 2 : Since there are three pairs of vertices with cost two and since they do not form the cycles with the previous edges we join the vertices (e, h) , (f, g) and (e, b) .



Step 3 : There are three pairs of vertices with the cost 3. They are (f, b) , (f, i) and (b, a) . If we join (f, i) it will form a cycle. We reject it and join the remaining two.



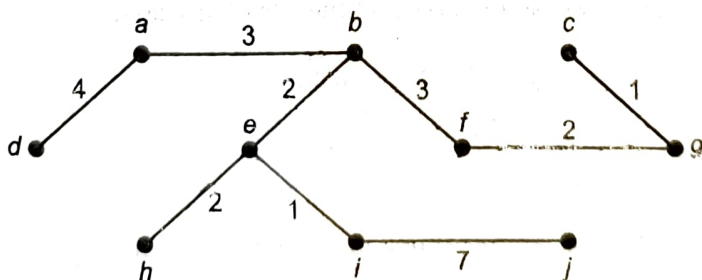
Step 4 : There are three vertices with the cost 4. They are (e, a) , (h, i) , (a, d) . We cannot join (e, a) and (h, i) as then they will form cycles. Hence, we join (a, d) .



Step 5 : We cannot join the next pair of vertices (d, e) with cost 5, as it will form a cycle.

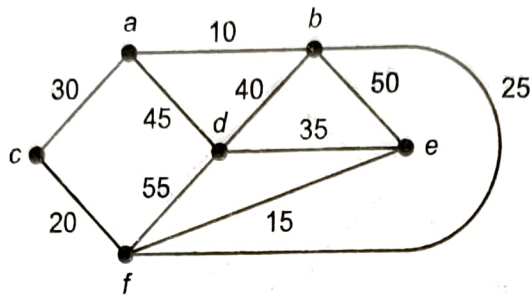
Step 6 : There are two pairs with cost 6. They are (d, h) and (f, c) . We reject both as they form cycles.

Step 7 : The next pair with cost 7 is (i, j) . We join it.



Since all the vertices are joined. This is the required minimum spanning tree with the same cost 25.

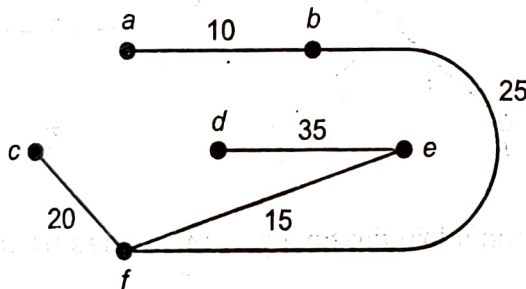
Example 6 : Find the minimum spanning tree by Kruskal's algorithm.
(M.U. 2005)



Sol. : We first list the edges in the increasing order of their costs

- (1) (a, b) (2) (f, e) (3) (c, f) (4) (f, b) (5) (a, c)
 (6) (d, e) (7) (d, b) (8) (a, d) (9) (b, e) (10) (d, f)

We now jot down the vertices along with their names.



Step 1 : We first join the pair with minimum cost 10 (a, b).

Step 2 : We then join the pair with the next least cost 15 (f, e).

Step 3 : We then join the pair with the next least cost 20 (c, f).

Step 4 : We then join the pair with the next least cost 25 (f, b).

Step 5 : The next pair with the least cost is (a, c).

But if we join these vertices, it will create a cycle. Hence, we move on to the next least cost pair.

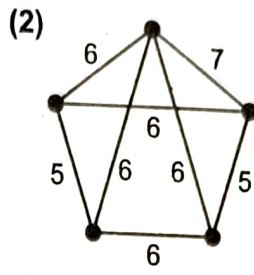
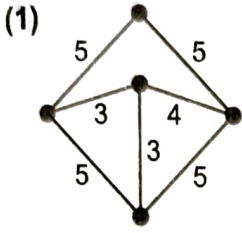
Step 6 : The next least cost pair is (d, e). We join these vertices.

Step 7 : Since all vertices are connected. This is the required minimum spanning tree.

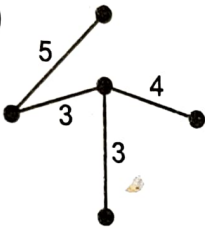
The cost is $10 + 15 + 20 + 25 + 35 = 105$.

EXERCISE - II

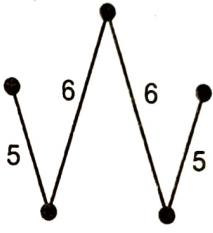
1. Find the minimal spanning tree for each of the following graphs.



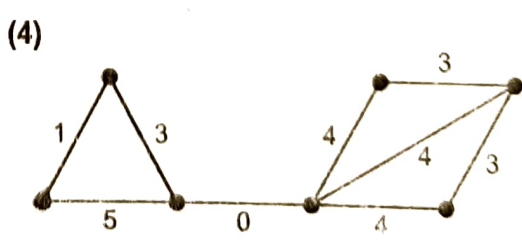
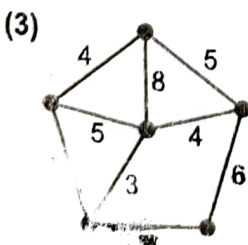
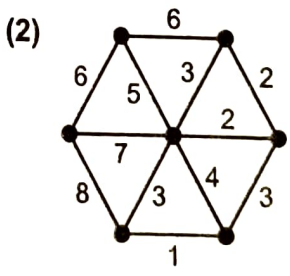
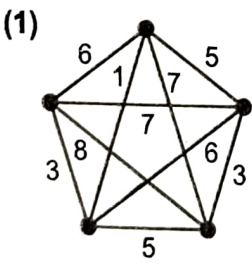
[Ans. : (1)



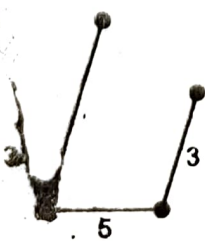
(2)



2. Obtain the minimal spanning tree and its weight for the following graphs.



[Ans. : (1)



(2)

